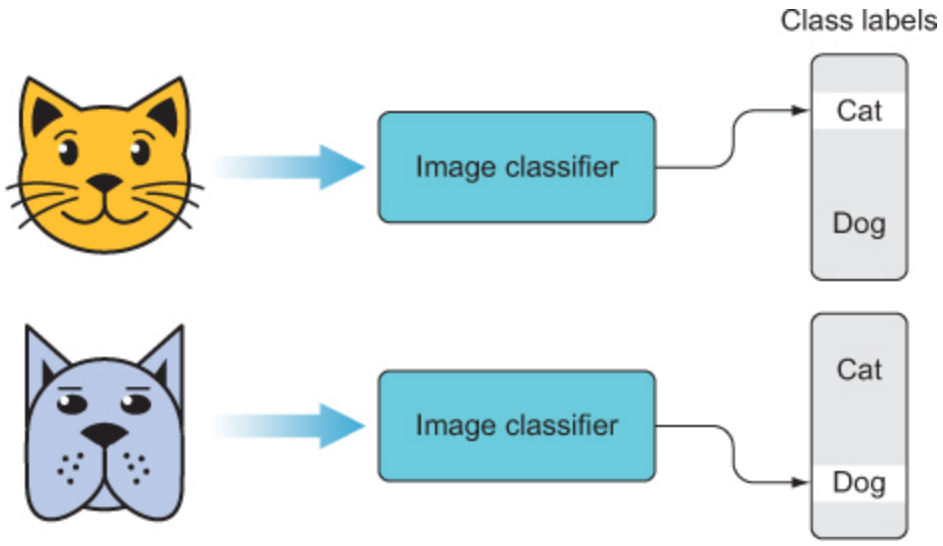


1.1. The “deep” in deep reinforcement learning

Deep learning models are just one of many kinds of machine learning models we can use to classify images. In general, we just need some sort of function that takes in an image and returns a class label (in this case, the label identifying which kind of animal is depicted in the image), and usually this function has a fixed set of adjustable *parameters*—we call these kinds of models *parametric* models. We start with a parametric model whose parameters are initialized to random values—this will produce random class labels for the input images. Then we use a *training* procedure to adjust the parameters so the function iteratively gets better and better at correctly classifying the images. At some point, the parameters will be at an optimal set of values, meaning that the model cannot get any better at the classification task. Parametric models can also be used for *regression*, where we try to fit a model to a set of data so we can make predictions for unseen data (figure 1.2). A more sophisticated approach might perform even better if it had more parameters or a better internal architecture.

Figure 1.2. Perhaps the simplest machine learning model is a simple linear function of the form *f(x) = mx + b*, with parameters *m* (the slope) and *b* (the intercept). Since it has adjustable parameters, we call it a parametric function or model. If we have some 2-dimensional data, we can start with a randomly initialized set of parameters, such as [m = 3.4, b = 0.3], and then use a training algorithm to optimize the parameters to fit the training data, in which case the optimal set of parameters is close to [m = 2, b = 1].